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标题	Original Fluorinated Non-Isocyanate Polyhydroxyurethanes
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摘要（原文节选）

Highlights What are the main findings? Fluorinated telechelic bis(cyclocarbonate) was synthesized through carbonylation of a fluorinated diepoxide.Fluorinated polyhydroxyurethanes (FPHUs) with different molar masses were synthesized through polyaddition reaction.Analysis showed that the FPHUs have higher molar mass, glass transition, and decomposition temperatures compared to their hydrogenated homologues. What is the implication of the main finding? New non-isocyanate polymers with improved properties.The synthesi …

数据表 · Table 2（15 行，含数值行 15 行）

Model reaction of (A) with hexylamine a: conditions optimization.

Entry	Solvent	(A): Amine	T (° C)	t (h)	Cat.(2 mol%)	Conv.(%) b	Yield (%) c	Secondary Alcohol (%) d
1	DMSO	1:2.0	25	48	-	55	48 e	66
2	DMSO	1:2.0	60	18	-	72	57 e	71
3	DMSO	1:2.0	80	18	-	81	55 e	70
4	DMF	1:2.0	80	18	-	92	49 e	73
5	Acetonitrile	1:2.0	80	18	-	79	52	65
6	Dimethyl carbonate	1:2.0	80	18	-	73	52	67
7	1,4-dioxane	1:2.0	80	18	-	85	48	70
8	Ethyl acetate	1:2.0	80	18	-	88	42	65
9	Trifluoro-toluene	1:2.0	80	18	-	56	50	68
10	-	1:2.0	80	18	-	59	48	65
11	-	1:2.6	80	18	-	70	64	60
12	-	1:3.0	80	18	-	100	69	65
13	-	1:2.0	80	5	NEt3	100	63	65
14	-	1:2.6	80	5	NEt3	100	84	68
15	-	1:3.0	80	5	NEt3	100	92	70

脚注（逐字）：a Conditions: carried out in a 10 mL flask; (A)/solvent (if any) 0.5 g/2 mL. b Determined by 1H NMR: Conv. % =(∫ 7.2NH (7,7 ′)2(∫ 7.2NH (7,7 ′)/2+ ∫ 4.6OCH2 (1,1 ′ of carbonate)/2 × 100; where ∫ iXY stands for the integral of the signal centered at i ppm assigned to XY group (Figure 5 and Figure S1 in the S

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数据表 · Table 3（4 行，含数值行 4 行）

FPHUs and PHUs synthesis from (A) or (B) with hexamethylenediamine a.

Run#	CC	A: Diamine	Conv.(%)	Yield(%) b	Secondary Alcohol (%) c	Primary Alcohol (%) c
FPHU P1	A	1:1.5	100	82	78	22
FPHU P2	A	1:1.3	100	66	71	29
FPHU P3 d	A	1:1.0	94	61	70	30
PHU P4 d	B	1:1.0	88	57	68	32

脚注（逐字）：a Conditions: Bulk polymerization, 2% of NEt3, 80 ° C, 5 h. b The yield was calculated after purification, and the residual excess diamine was removed at the end of the reaction by distillation under a vacuum. c The proportions of primary and secondary alcohols were calculated using 1H NMR spectra of

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